Mathematical Statistics

Practice problems for the SNU Statistics graduate entrance exam

12 problems × 10 points. Solutions start on page 5.

Notation: $\mathrm{Exp}(\theta)$ has mean θ; $\mathrm{Gamma}(\alpha,\beta)$ has shape α and scale β; $\chi^2_\alpha(r)$, $t_\alpha(r)$, $F_\alpha(r_1,r_2)$, z_α are upper-α points. A sequence Y_n is $O_p(1)$ if for every $\epsilon>0$ there are M,N with $P(|Y_n|>M)<\epsilon$ for all $n\ge N$; $Y_n=o_p(1)$ means $Y_n \xrightarrow{p} 0$; $Y_n=O_p(a_n)$ means $Y_n/a_n=O_p(1)$.

Problems

Problem 1 *Analysis* [10 pts]

(a) Using the mean value theorem, show that $\dfrac{x}{1+x} < \log(1+x) < x$ for all $x>0$.

(b) Let $f_n(x)=nxe^{-nx}$ for $x\ge 0$. Find the pointwise limit f. Is the convergence uniform on $[0,\infty)$? On $[a,\infty)$ for a fixed $a>0$?

(c) Show that $\int_0^1 f_n(x)\,dx \to \int_0^1 f(x)\,dx$, even though the convergence in (b) is not uniform on $[0,1]$.

(d) Let $K\subset\mathbb{R}^k$ be compact and $g:K\to\mathbb{R}$ continuous. Prove that g is uniformly continuous on K.

Problem 2 *Joint, marginal and conditional distributions* [10 pts]

(X,Y) is uniformly distributed on $D=\{(x,y): 0<x<1,\ 0<y<x^2\}$.

(a) Find the joint pdf and the marginal pdfs of X and Y.

(b) Find $E(Y\mid X=x)$ and $\mathrm{Var}(Y\mid X=x)$, and use them to compute EY and $\mathrm{Var}\,Y$.

(c) Compute $\mathrm{Cov}(X,Y)$. Are X and Y independent?

(d) Compute $P(Y>X/2)$.

Problem 3 *Transformations* [10 pts]

Let $X,Y \overset{\text{iid}}{\sim} \mathrm{Exp}(1)$.

(a) Find the joint pdf of $U=X/(X+Y)$ and $V=X+Y$. Show that they are independent and identify their distributions.

(b) Using (a), find $E(X\mid X+Y=v)$ and $E\big(X^2\mid X+Y=v\big)$.

(c) Find the pdf of $W=X-Y$.

(d) Let $m=\min(X,Y)$ and $R=\max(X,Y)-\min(X,Y)$. Show that m and R are independent and find their distributions.

Problem 4 *Multivariate normal* [10 pts]

Let $\mathbf{X}=(X_1,X_2,X_3)'\sim N_3(\mathbf{0},\Sigma)$ with $\Sigma=\begin{pmatrix}2&1&0\\1&2&1\\0&1&2\end{pmatrix}$.

(a) Find the distribution of $X_1 - X_2 + X_3$.

(b) Find the constant c for which $X_2 - c(X_1 + X_3)$ and $X_1 + X_3$ are independent, and justify independence.

(c) Find the conditional distribution of X_2 given $(X_1, X_3) = (x_1, x_3)$.

(d) Find the mgf $M(t) = E\exp(tX_1X_3)$ and state where it is finite.

(e) What is the distribution of $\mathbf{X}'\Sigma^{-1}\mathbf{X}$? Justify.

Problem 5 *Stochastic order and limits* **[10 pts]**

(a) Show that if $X_n \xrightarrow{d} X$, then $X_n = O_p(1)$.

(b) Show that if $X_n = O_p(1)$ and $Y_n = o_p(1)$, then $X_nY_n = o_p(1)$.

(c) Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} U(0, \theta)$. Show that $n(\theta - X_{(n)}) \xrightarrow{d} \text{Exp}(\theta)$. Conclude that $X_{(n)} - \theta = O_p(n^{-1})$ and $\sqrt{n}(X_{(n)} - \theta) \xrightarrow{p} 0$.

(d) Let $X_1, \dots, X_n$ be i.i.d. with mean μ and variance $\sigma^2 < \infty$. Without quoting the delta method, show $\sqrt{n}(\bar{X}_n^2 - \mu^2) \xrightarrow{d} N(0, 4\mu^2\sigma^2)$, using the decomposition $\bar{X}_n^2 - \mu^2 = 2\mu(\bar{X}_n - \mu) + (\bar{X}_n - \mu)^2$.

Problem 6 *MLE, MME and efficiency* **[10 pts]**

Let $X_1, \dots, X_n$ be a sample from the Rayleigh density

$$f(x; \theta) = \frac{x}{\theta}\exp\Big(-\frac{x^2}{2\theta}\Big), \qquad x > 0,\ \theta > 0.$$

(a) Show that $X_1^2 \sim \text{Exp}(2\theta)$. Find the MLE $\hat{\theta}$, and its mean and variance.

(b) Compute the Fisher information $I(\theta)$ and show that $\hat{\theta}$ attains the Cramér–Rao lower bound. Is $\hat{\theta}$ the UMVUE?

(c) Using $EX_1 = \sqrt{\pi\theta/2}$, find the method-of-moments estimator $\tilde{\theta}$ based on $\bar{X}$ and its limiting distribution.

(d) Compute $ARE(\tilde{\theta}, \hat{\theta})$.

Problem 7 *Multinomial reparametrization* **[10 pts]**

A sample of n individuals is classified into three genotypes with counts

$$(N_1, N_2, N_3) \sim \text{Multi}\big(n;\ \theta^2,\ 2\theta(1-\theta),\ (1-\theta)^2\big), \qquad 0 < \theta < 1.$$

(a) Find the MLE $\hat{\theta}$.

(b) Show that $2N_1 + N_2 \sim B(2n, \theta)$, and find $E\hat{\theta}$ and $\text{Var}\,\hat{\theta}$.

(c) Find the Fisher information per individual and the limiting distribution of $\sqrt{n}(\hat{\theta} - \theta)$. Is $\hat{\theta}$ the UMVUE?

(d) With $n = 200$ and $(N_1, N_2, N_3) = (50, 90, 60)$, test at level 0.05 whether the three-cell model above fits, against a general trinomial, using Pearson's statistic. State the degrees of freedom.

Problem 8 *Sufficiency, completeness, UMVUE* **[10 pts]**

Let $X_1, \dots, X_n$ be a sample from $f(x;\theta) = e^{-(x-\theta)}$, $x > \theta$, $\theta \in \mathbb{R}$.

(a) Show that $X_{(1)}$ is sufficient, and find its distribution.

(b) Show that $X_{(1)}$ is complete.

(c) Find the UMVUEs of θ and of θ^2.

(d) Show that $X_{(1)}$ and $S = \sum_{i=1}^{n}(X_i - X_{(1)})$ are independent, and find the distribution of S.

Problem 9 *Most powerful and UMP tests* **[10 pts]**

Let $X_1, \dots, X_n$ be a sample from the Pareto density $f(x;\theta) = \theta x^{-\theta-1}$, $x > 1$, $\theta > 0$, and let $T = \sum_{i=1}^{n} \log X_i$.

(a) Show that $2\theta T \sim \chi^2(2n)$.

(b) Find the most powerful level-α test of $H_0 : \theta = 1$ vs $H_1 : \theta = \theta_1$ for a fixed $\theta_1 > 1$.

(c) Show that the test in (b) is UMP for $H_0 : \theta \le 1$ vs $H_1 : \theta > 1$, and express its power function with a chi-square cdf.

(d) For $n = 10$ and $\alpha = 0.05$ ($\chi^2_{0.95}(20) = 10.851$), give the rejection region and the power at $\theta = 2$.

Problem 10 *Likelihood ratio test* **[10 pts]**

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Exp}(\theta_1)$ and $Y_1, \dots, Y_m \overset{\text{iid}}{\sim} \text{Exp}(\theta_2)$ be independent samples. Consider $H_0 : \theta_1 = \theta_2$ vs $H_1 : \theta_1 \neq \theta_2$.

(a) Show that $2\log\Lambda$, the LRT statistic, is a function of $B = n\bar{X}/(n\bar{X} + m\bar{Y})$, and describe the form of the rejection region.

(b) Find the exact null distribution of B, and of $\bar{X}/\bar{Y}$.

(c) State the limiting null distribution of $2\log\Lambda$ as $n, m \to \infty$.

Problem 11 *Signed rank test* **[10 pts]**

Let $X_1, \dots, X_n$ be a sample from a continuous distribution symmetric about θ, and $W^+ = \sum_i \mathbf{1}\{X_i > 0\} R_i$, where R_i is the rank of $|X_i|$ among $|X_1|, \dots, |X_n|$. Under $H_0 : \theta = 0$:

(a) Show that $W^+ \overset{d}{=} \sum_{j=1}^{n} j B_j$ with $B_j \overset{\text{iid}}{\sim} \text{Bernoulli}(1/2)$. (You may use that the signs and the ranks of $|X_i|$ are independent.)

(b) Find EW^+ and $\operatorname{Var} W^+$, and the exact null distribution of W^+ for $n = 4$.

(c) Show that $W^+ = \sum_{1 \le i \le j \le n} \mathbf{1}\{X_i + X_j > 0\}$.

(d) The ARE of the signed rank test relative to the t test is $12\sigma^2\big(\int f^2\big)^2$, where f is the density of $X_1 - \theta$ and σ^2 its variance. Evaluate it for f uniform on $(-1, 1)$ and for $f(x) = \tfrac{1}{2}e^{-|x|}$.

Problem 12 *Bayes and minimax* **[10 pts]**

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$ with prior $\theta \sim \text{Exp}(\beta)$, i.e. Gamma$(1, \beta)$. Let $T = \sum X_i$ and $s = (n + 1/\beta)^{-1}$.

(a) Find the posterior distribution of θ.

(b) Find the Bayes estimator under squared error loss, and write it as a weighted average of $\bar{X}$ and the prior mean.

(c) Under the loss $L(\theta, a) = (a - \theta)^2/\theta$, show that the Bayes estimator is $a^* = sT$ and that its posterior expected loss is s for every T.

(d) Show that $\bar{X}$ is minimax for the loss in (c).

Solutions

Solution 1

(a) Apply the MVT to $\log(1+t)$ on $[0,x]$: $\log(1+x) = \frac{x}{1+\xi}$ for some $\xi \in (0,x)$. Since $1 < 1+\xi < 1+x$, we get $\frac{x}{1+x} < \log(1+x) < x$.

(b) $f_n(0) = 0$, and for $x > 0$, $nxe^{-nx} \to 0$ because $ue^{-u} \to 0$ as $u \to \infty$. So $f \equiv 0$. Since $\frac{d}{dx} f_n = ne^{-nx}(1-nx)$, f_n peaks at $x = 1/n$ with $\sup_{x\ge 0} f_n = f_n(1/n) = e^{-1} \not\to 0$: not uniform on $[0,\infty)$. On $[a,\infty)$, for $n > 1/a$ the function is decreasing there, so $\sup_{x\ge a} f_n = nae^{-na} \to 0$: uniform.

(c) $\int_0^1 nxe^{-nx}dx = \frac{1}{n}\int_0^n ue^{-u}du \le \frac{1}{n}\int_0^\infty ue^{-u}du = \frac{1}{n} \to 0 = \int_0^1 f$.

(d) Suppose not. Then there are $\epsilon > 0$ and $x_n, y_n \in K$ with $|x_n - y_n| < 1/n$ but $|g(x_n) - g(y_n)| \ge \epsilon$. By compactness (Bolzano–Weierstrass), a subsequence $x_{n_k} \to x \in K$, and then $y_{n_k} \to x$ as well. By continuity $g(x_{n_k}) - g(y_{n_k}) \to g(x) - g(x) = 0$, contradicting $\ge \epsilon$.

Solution 2

(a) The area of D is $\int_0^1 x^2 dx = \frac{1}{3}$, so $f(x,y) = 3$ on D. Then $f_X(x) = \int_0^{x^2} 3\,dy = 3x^2$ on $(0,1)$, and $f_Y(y) = \int_{\sqrt{y}}^1 3\,dx = 3(1-\sqrt{y})$ on $(0,1)$.

(b) Given $X = x$, $Y \sim U(0,x^2)$: $E(Y \mid x) = x^2/2$, $\operatorname{Var}(Y \mid x) = x^4/12$. Then $EY = E(X^2)/2 = \frac{1}{2}\cdot\frac{3}{5} = \frac{3}{10}$, and

$$\operatorname{Var} Y = E\Big(\frac{X^4}{12}\Big) + \operatorname{Var}\Big(\frac{X^2}{2}\Big) = \frac{1}{12}\cdot\frac{3}{7} + \frac{1}{4}\Big(\frac{3}{7} - \frac{9}{25}\Big) = \frac{1}{28} + \frac{3}{175} = \frac{37}{700}.$$

(c) $E(XY) = E\{X\,E(Y \mid X)\} = E(X^3)/2 = \frac{1}{2}\cdot\frac{3}{6} = \frac{1}{4}$, $EX = \frac{3}{4}$, so $\operatorname{Cov}(X,Y) = \frac{1}{4} - \frac{3}{4}\cdot\frac{3}{10} = \frac{1}{40}$. Not independent (nonzero covariance; also the support is not a product set).

(d) $\{y > x/2\} \cap D$ is nonempty only where $x^2 > x/2$, i.e. $x > 1/2$:

$$P(Y > X/2) = \int_{1/2}^1 3\Big(x^2 - \frac{x}{2}\Big)dx = 3\Big[\frac{x^3}{3} - \frac{x^2}{4}\Big]_{1/2}^1 = 3\Big(\frac{1}{12} + \frac{1}{48}\Big) = \frac{5}{16}.$$

Solution 3

(a) Inverse map $x = uv$, $y = (1-u)v$, $0 < u < 1$, $v > 0$, with $|J| = |\det \begin{pmatrix} v & u \\ -v & 1-u \end{pmatrix}| = v$. So $f(u,v) = e^{-v}v = \mathbf{1}_{(0,1)}(u)\cdot ve^{-v}$: $U \sim U(0,1)$ and $V \sim \text{Gamma}(2,1)$, independent.

(b) $X = UV$ with U independent of V, so $E(X \mid V = v) = vEU = v/2$ and $E(X^2 \mid V = v) = v^2 EU^2 = v^2/3$.

(c) For $w > 0$, $f_W(w) = \int_0^\infty f_X(y+w)f_Y(y)\,dy = \int_0^\infty e^{-(y+w)}e^{-y}dy = \frac{1}{2}e^{-w}$; by symmetry $f_W(w) = \frac{1}{2}e^{-|w|}$, $w \in \mathbb{R}$ (double exponential).

(d) The order statistics have joint pdf $2e^{-y_1-y_2}$ on $0 < y_1 < y_2$. With $m = y_1$, $r = y_2 - y_1$ ($|J| = 1$), $f(m,r) = 2e^{-2m-r} = 2e^{-2m}\cdot e^{-r}$ on $m, r > 0$. So $m \sim \text{Exp}(1/2)$ and $R \sim \text{Exp}(1)$ are independent (memorylessness).

Solution 4

(a) With $a = (1,-1,1)'$, $\Sigma a = (1,0,1)'$ and $a'\Sigma a = 2$. So $X_1 - X_2 + X_3 \sim N(0,2)$.

(b) $(X_2 - c(X_1+X_3),\ X_1+X_3)$ is a linear image of $\mathbf{X}$, hence bivariate normal, and is independent iff the covariance is 0. $\operatorname{Cov}(X_2, X_1+X_3) = 1+1 = 2$ and $\operatorname{Var}(X_1+X_3) = 2+2+2\cdot 0 = 4$, so $\operatorname{Cov} = 2 - 4c = 0$ gives $c = \frac{1}{2}$.

(c) $\Sigma_{12} = (1,1)$, $\Sigma_{22} = 2I_2$, so $\Sigma_{12}\Sigma_{22}^{-1} = (\frac{1}{2},\frac{1}{2})$. Hence $X_2 \mid (x_1,x_3) \sim N\big(\frac{x_1+x_3}{2},\ 2 - \frac{1}{2}(1+1)\big) = N\big(\frac{x_1+x_3}{2}, 1\big)$.

(d) $\mathrm{Cov}(X_1, X_3) = 0$ and they are jointly normal, so $X_1, X_3 \overset{\text{iid}}{\sim} N(0,2)$. Conditioning on X_3, $E(e^{tX_1X_3} \mid X_3) = \exp(\frac{1}{2} \cdot 2t^2X_3^2) = e^{t^2X_3^2}$. Since $X_3^2/2 \sim \chi^2(1)$, $Ee^{t^2X_3^2} = Ee^{2t^2(X_3^2/2)} = (1-4t^2)^{-1/2}$. So $M(t) = (1-4t^2)^{-1/2}$ for $|t| < \frac{1}{2}$, and $M(t) = \infty$ otherwise.

(e) Σ is positive definite (leading minors $2, 3, 4$). $\mathbf{Z} = \Sigma^{-1/2}\mathbf{X} \sim N_3(\mathbf{0}, I)$ and $\mathbf{X}'\Sigma^{-1}\mathbf{X} = \mathbf{Z}'\mathbf{Z} \sim \chi^2(3)$.

Solution 5

(a) Let F be the cdf of X and $\epsilon > 0$. Since F has at most countably many discontinuities, choose M such that $\pm M$ are continuity points and $P(X > M) + P(X \le -M) < \epsilon/2$. Then

$$P(|X_n| > M) \le 1 - F_n(M) + F_n(-M) \to 1 - F(M) + F(-M) < \epsilon/2,$$

so $P(|X_n| > M) < \epsilon$ for all large n; the finitely many earlier n are handled by enlarging M.

(b) For $\delta, \epsilon > 0$, choose M with $P(|X_n| > M) < \epsilon$ for large n. Then

$$P(|X_nY_n| > \delta) \le P(|X_n| > M) + P(|Y_n| > \delta/M) < \epsilon + o(1).$$

(c) For $0 \le x \le n\theta$, $P(n(\theta - X_{(n)}) > x) = P(X_{(n)} < \theta - x/n) = (1 - \frac{x}{n\theta})^n \to e^{-x/\theta}$. So $n(\theta - X_{(n)}) \xrightarrow{d} \mathrm{Exp}(\theta)$, which is $O_p(1)$ by (a), i.e. $X_{(n)} - \theta = O_p(n^{-1})$. Then $\sqrt{n}(X_{(n)} - \theta) = n^{-1/2} \cdot n(X_{(n)} - \theta) = o(1)O_p(1) = o_p(1)$ by (b).

(d) By the CLT, $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$, so $2\mu\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, 4\mu^2\sigma^2)$. Also $\sqrt{n}(\bar{X}_n - \mu)^2 = \sqrt{n}(\bar{X}_n - \mu) \cdot (\bar{X}_n - \mu) = O_p(1)\, o_p(1) = o_p(1)$, by (a), the WLLN and (b). Slutsky's theorem gives the result.

Solution 6

(a) $P(X^2 > t) = \int_{\sqrt{t}}^{\infty} \frac{x}{\theta} e^{-x^2/2\theta} dx = e^{-t/(2\theta)}$, so $X^2 \sim \mathrm{Exp}(2\theta)$. The log-likelihood is $l(\theta) = \sum \log x_i - n \log\theta - \sum x_i^2/(2\theta)$, with $\dot{l} = -n/\theta + \sum x_i^2/(2\theta^2)$ positive iff $\theta < \sum x_i^2/(2n)$. So $\hat{\theta} = \frac{1}{2n}\sum X_i^2$, with $E\hat{\theta} = \theta$ and $\mathrm{Var}\,\hat{\theta} = \frac{(2\theta)^2}{4n} = \frac{\theta^2}{n}$.

(b) $\partial_\theta^2 \log f = \frac{1}{\theta^2} - \frac{x^2}{\theta^3}$, so $I(\theta) = -\frac{1}{\theta^2} + \frac{2\theta}{\theta^3} = \frac{1}{\theta^2}$. The CR bound for unbiased estimators is $\frac{1}{nI(\theta)} = \frac{\theta^2}{n} = \mathrm{Var}\,\hat{\theta}$. So $\hat{\theta}$ is efficient, hence UMVUE (also: $\sum X_i^2$ is complete sufficient in this exponential family).

(c) $\theta = \frac{2}{\pi}(EX)^2$, so $\tilde{\theta} = \frac{2}{\pi}\bar{X}^2$. $\mathrm{Var}\,X = EX^2 - (EX)^2 = 2\theta - \frac{\pi\theta}{2} = \frac{(4-\pi)\theta}{2}$. With $g(m) = 2m^2/\pi$, $g'(m) = 4m/\pi$ and $m = \sqrt{\pi\theta/2}$, the delta method gives

$$\sqrt{n}(\tilde{\theta} - \theta) \xrightarrow{d} N\Big(0, \frac{16m^2}{\pi^2} \cdot \frac{(4-\pi)\theta}{2}\Big) = N\Big(0, \frac{4(4-\pi)}{\pi}\theta^2\Big).$$

(d) $ARE(\tilde{\theta}, \hat{\theta}) = \dfrac{\theta^2}{4(4-\pi)\theta^2/\pi} = \dfrac{\pi}{4(4-\pi)} \approx 0.915.$

Solution 7

(a) $l(\theta) = (2N_1 + N_2)\log\theta + (N_2 + 2N_3)\log(1-\theta) + \text{const}$, strictly concave, so $\hat{\theta} = \frac{2N_1+N_2}{2n}$.

(b) Each individual contributes $Y_i \in \{2, 1, 0\}$ copies of the allele, with probabilities $\theta^2, 2\theta(1-\theta), (1-\theta)^2$, i.e. $Y_i \sim B(2, \theta)$, independently. So $2N_1 + N_2 = \sum Y_i \sim B(2n, \theta)$, $E\hat{\theta} = \theta$, $\mathrm{Var}\,\hat{\theta} = \frac{\theta(1-\theta)}{2n}$.

(c) Per individual, $I(\theta) = \mathrm{Var}\left(\frac{Y}{\theta} - \frac{2-Y}{1-\theta}\right) = \frac{2\theta(1-\theta)}{\theta^2(1-\theta)^2} = \frac{2}{\theta(1-\theta)}$. Hence $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \theta(1-\theta)/2)$ (here even $\mathrm{Var}\,\hat{\theta} = 1/(nI)$ exactly). $2N_1 + N_2$ is complete sufficient (binomial family), and $\hat{\theta}$ is an unbiased function of it: UMVUE.

(d) $\hat{\theta} = (100 + 90)/400 = 0.475$. Expected counts: $200(0.475)^2 = 45.125$, $200 \cdot 2(0.475)(0.525) = 99.75$, $200(0.525)^2 = 55.125$. Pearson's statistic:

$$\frac{4.875^2}{45.125} + \frac{9.75^2}{99.75} + \frac{4.875^2}{55.125} = 0.527 + 0.953 + 0.431 = 1.91.$$

Degrees of freedom $= \dim\Omega - \dim\Omega_0 = 2-1 = 1$; since $1.91 < \chi^2_{0.05}(1) = 3.84$, the model is not rejected.

Solution 8

(a) $\prod f(x_i;\theta) = e^{-\sum x_i + n\theta}\mathbf{1}\{x_{(1)} > \theta\} = \big[e^{n\theta}\mathbf{1}\{x_{(1)} > \theta\}\big]e^{-\sum x_i}$; by factorization $X_{(1)}$ is sufficient. $P(X_{(1)} > y) = e^{-n(y-\theta)}$ for $y > \theta$, so $X_{(1)} - \theta \sim \mathrm{Exp}(1/n)$.

(b) If $E_\theta g(X_{(1)}) = \int_\theta^\infty g(y) n e^{-n(y-\theta)}dy = 0$ for all θ, then $\int_\theta^\infty g(y)e^{-ny}dy = 0$ for all θ. Differentiating in θ, $g(\theta)e^{-n\theta} = 0$ for almost every θ, so $g = 0$ a.e. and $P_\theta(g(X_{(1)}) = 0) = 1$.

(c) $EX_{(1)} = \theta + \frac{1}{n}$, so $X_{(1)} - \frac{1}{n}$ is the UMVUE of θ (Lehmann–Scheffé). Also $EX^2_{(1)} = \mathrm{Var}\, X_{(1)} + (EX_{(1)})^2 = \frac{1}{n^2} + \theta^2 + \frac{2\theta}{n} + \frac{1}{n^2}$, so

$$E\Big[X^2_{(1)} - \frac{2}{n}X_{(1)}\Big] = \theta^2 + \frac{2\theta}{n} + \frac{2}{n^2} - \frac{2\theta}{n} - \frac{2}{n^2} = \theta^2,$$

and $X^2_{(1)} - \frac{2}{n}X_{(1)}$ is the UMVUE of θ^2.

(d) Write $X_i = \theta + Z_i$ with $Z_i \overset{\text{iid}}{\sim} \mathrm{Exp}(1)$; then $S = \sum(Z_i - Z_{(1)})$ does not depend on θ: it is ancillary. $X_{(1)}$ is complete sufficient, so by Basu's theorem they are independent. With $D_i = (n-i+1)(Z_{(i)} - Z_{(i-1)}) \overset{\text{iid}}{\sim} \mathrm{Exp}(1)$ (exponential spacings), $S = \sum_{i=2}^n (Z_{(i)} - Z_{(1)}) = \sum_{i=2}^n D_i \sim \mathrm{Gamma}(n-1, 1)$.

Solution 9

(a) $P(\log X > t) = P(X > e^t) = e^{-\theta t}$, so $\log X_i \overset{\text{iid}}{\sim} \mathrm{Exp}(1/\theta)$, $T \sim \mathrm{Gamma}(n, 1/\theta)$ and $2\theta T \sim \mathrm{Gamma}(n, 2) = \chi^2(2n)$.

(b) $\dfrac{\prod f(x_i;\theta_1)}{\prod f(x_i;1)} = \theta_1^n \exp\{-(\theta_1 - 1)T\}$, decreasing in T for $\theta_1 > 1$. By Neyman–Pearson, the MP test rejects for $T \le c$. Under $\theta = 1$, $2T \sim \chi^2(2n)$, so $c = \frac{1}{2}\chi^2_{1-\alpha}(2n)$: reject when $2T \le \chi^2_{1-\alpha}(2n)$.

(c) The test does not depend on θ_1, so it is UMP for $\theta = 1$ vs $\theta > 1$. Its power is

$$\gamma(\theta) = P_\theta\big(2T \le \chi^2_{1-\alpha}(2n)\big) = P\big(\chi^2(2n) \le \theta\chi^2_{1-\alpha}(2n)\big),$$

increasing in θ. So $\max_{\theta\le 1}\gamma(\theta) = \gamma(1) = \alpha$, and any level-$\alpha$ test of $\theta \le 1$ is level α for $\theta = 1$; hence the test is UMP for $H_0 : \theta \le 1$. (This is also the one-parameter exponential family result with $g(\theta) = -\theta$ decreasing in θ, applied to T.)

(d) Reject when $2T \le 10.851$, i.e. $\sum \log X_i \le 5.43$. Power at $\theta = 2$: $P(\chi^2(20) \le 21.70) \approx 0.64$.

Solution 10

(a) Unrestricted MLEs: $\hat\theta_1 = \bar X$, $\hat\theta_2 = \bar Y$; under H_0: $\hat\theta_0 = (n\bar X + m\bar Y)/(n+m)$. Since $l(\theta_1,\theta_2) = -n\log\theta_1 - n\bar x/\theta_1 - m\log\theta_2 - m\bar y/\theta_2$, the terms $n\bar x/\hat\theta$ etc. sum to $n+m$ in both cases, and

$$2\log\Lambda = 2\big\{(n+m)\log\hat\theta_0 - n\log\bar X - m\log\bar Y\big\}.$$

With $B = n\bar X/(n\bar X + m\bar Y)$, $\bar X = \frac{(n+m)\hat\theta_0}{n}B$ and $\bar Y = \frac{(n+m)\hat\theta_0}{m}(1-B)$, so

$$2\log\Lambda = -2\{n\log B + m\log(1-B)\} + 2\{n\log\tfrac{n}{n+m} + m\log\tfrac{m}{n+m}\}.$$

The function $-n\log b - m\log(1-b)$ is convex with minimum at $b = n/(n+m)$, so the LRT rejects when $B \le c_1$ or $B \ge c_2$, where $c_1 < \frac{n}{n+m} < c_2$ give equal values of the statistic and total size α.

(b) Under H_0 (common θ), $n\bar X/\theta \sim \mathrm{Gamma}(n,1)$ and $m\bar Y/\theta \sim \mathrm{Gamma}(m,1)$ are independent, so $B \sim \mathrm{Beta}(n,m)$, free of θ. Equivalently $\frac{2n\bar X/\theta}{2n}\Big/\frac{2m\bar Y/\theta}{2m} = \bar X/\bar Y \sim F(2n,2m)$.

(c) $\dim\Omega - \dim\Omega_0 = 2-1 = 1$, so $2\log\Lambda \xrightarrow{d} \chi^2(1)$ under H_0.

Solution 11

(a) Let D_j be the index of the observation whose $|X|$ has rank j. Then $W^+ = \sum_{j=1}^{n} j\,\mathbf{1}\{X_{D_j} > 0\}$. Since the sign vector is independent of the ranks and its entries are i.i.d. fair, the signs attached to ranks $1, \dots, n$ are again i.i.d. fair, so $W^+ \overset{d}{=} \sum j B_j$.

(b) $EW^+ = \frac{1}{2}\sum j = \frac{n(n+1)}{4}$ and $\operatorname{Var} W^+ = \frac{1}{4}\sum j^2 = \frac{n(n+1)(2n+1)}{24}$. For $n = 4$, W^+ is the sum of a uniformly random subset of $\{1, 2, 3, 4\}$ (16 subsets):

w	0	1	2	3	4	5	6	7	8	9	10
$16\,P(W^+ = w)$	1	1	1	2	2	2	2	2	1	1	1

(c) Order by absolute value: $|Y_{(1)}| < \cdots < |Y_{(n)}|$. Then $W^+ = \sum_i i\mathbf{1}\{Y_{(i)} > 0\} = \sum_i \mathbf{1}\{Y_{(i)} > 0\} + \sum_{j<i} \mathbf{1}\{Y_{(i)} > 0\}$, and for $j < i$, $Y_{(i)} > 0 \iff Y_{(i)} + Y_{(j)} > 0$ since $|Y_{(j)}| < |Y_{(i)}|$. Hence $W^+ = \sum_{i \le j} \mathbf{1}\{X_i + X_j > 0\}$.

(d) Uniform on $(-1, 1)$: $\int f^2 = \frac{1}{2}$, $\sigma^2 = \frac{1}{3}$, so $ARE = 12 \cdot \frac{1}{3} \cdot \frac{1}{4} = 1$. Double exponential: $\int f^2 = \int \frac{1}{4}e^{-2|x|} = \frac{1}{4}$, $\sigma^2 = 2$, so $ARE = 12 \cdot 2 \cdot \frac{1}{16} = \frac{3}{2}$.

Solution 12

(a) $\pi(\theta \mid \mathbf{x}) \propto e^{-n\theta}\theta^T \cdot e^{-\theta/\beta} = \theta^T e^{-\theta/s}$, so $\theta \mid \mathbf{x} \sim \text{Gamma}(T + 1, s)$.

(b) The posterior mean is $s(T + 1) = \dfrac{n}{n + 1/\beta}\bar{X} + \dfrac{1/\beta}{n + 1/\beta}\beta$, a weighted average of $\bar{X}$ and the prior mean β.

(c) The posterior expected loss is $\rho(a) = a^2 E(\theta^{-1} \mid \mathbf{x}) - 2a + E(\theta \mid \mathbf{x})$. For $T \ge 1$, $E(\theta^{-1} \mid \mathbf{x}) = \frac{1}{sT}$, so ρ is minimized at $a^* = sT$ with $\rho(a^*) = s(T + 1) - sT = s$. For $T = 0$, $E(\theta^{-1} \mid \mathbf{x}) = \infty$, so only $a = 0 = sT$ has finite loss, $\rho(0) = E(\theta \mid \mathbf{x}) = s$. In all cases the Bayes estimator is sT and its posterior expected loss is s, so its Bayes risk is $r(\pi_\beta) = s = (n + 1/\beta)^{-1}$.

(d) $\bar{X}$ has risk $E\big[(\bar{X} - \theta)^2/\theta\big] = \frac{\theta/n}{\theta} = \frac{1}{n}$ for all θ. For any estimator δ, $\max_\theta R_\delta(\theta) \ge r(\pi_\beta, \delta) \ge (n + 1/\beta)^{-1}$ for every β; letting $\beta \to \infty$, $\max_\theta R_\delta \ge \frac{1}{n} = \max_\theta R_{\bar{X}}$. So $\bar{X}$ is minimax (limit of Bayes estimators, least favourable sequence of priors).